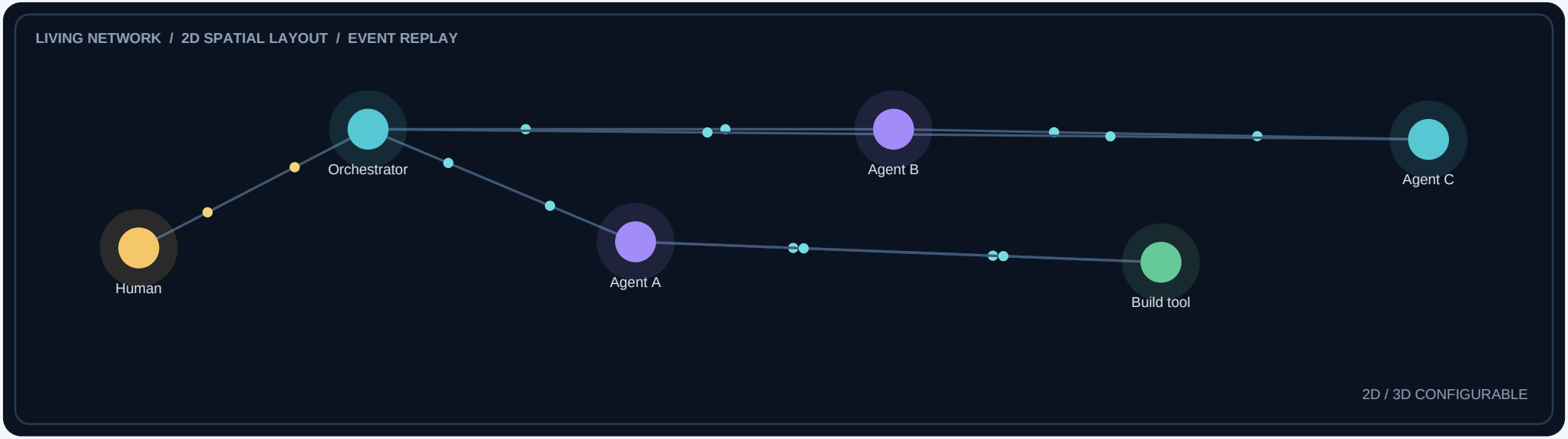
Living Network

A spatial, animated map of agents, tools, messages, and delegated work.



WHY THIS VIEW

Make the software factory feel alive. See which actors are active, how work is delegated, and where messages and results travel.

VISUAL MODEL

Agents, tools, and human participants occupy spatial nodes. Messages, delegations, and results travel between them as glowing particles or animated flows. The network supports configurable 2D and 3D layouts.

VISUAL ENCODING

Node color = actor type • node pulse = recent activity • flow direction = sender to recipient • flow density = message volume or tokens

KEY INTERACTIONS

Switch 2D/3D; pause and scrub replay; isolate an actor and its neighborhood; inspect a message flow; filter by event type; expand a subagent cluster.

DESIGN GUARDRAILS

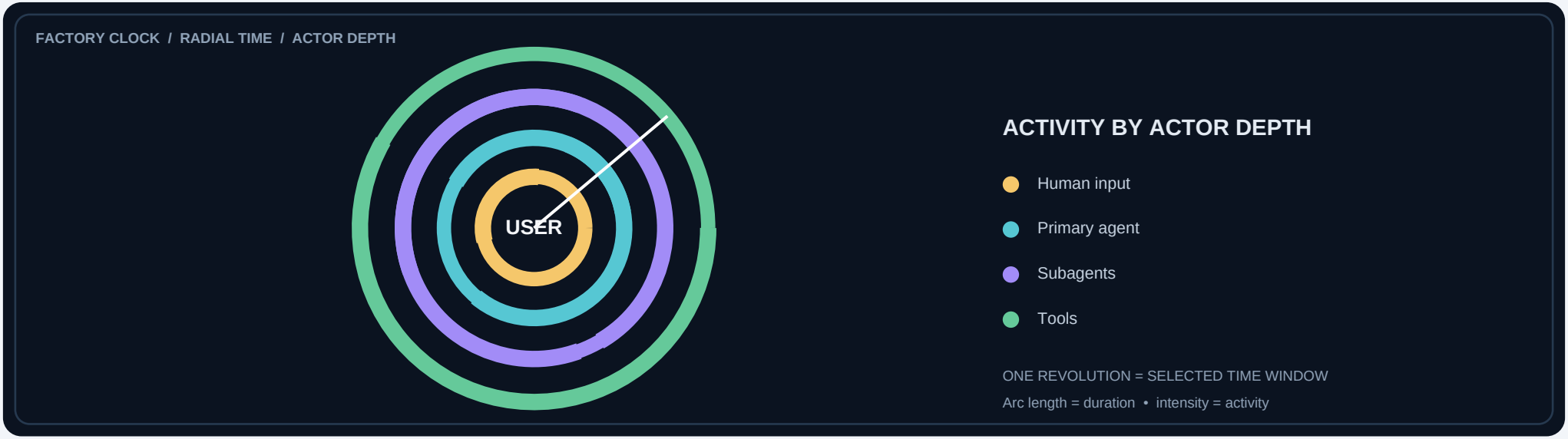
Prefer a stable layout with gentle adaptation. Aggregate repeated traffic and limit simultaneous particles. Clearly distinguish live activity from accelerated historical replay.

PRIMARY QUESTION

What is happening across the system, and how is work moving?

Factory Clock

A radial sweep of activity from the human at the center to agents and tools at the perimeter.



WHY THIS VIEW

Reveal when work happened, how long it lasted, and how activity was distributed across the agent hierarchy—including parallel work.

VISUAL MODEL

The angle represents elapsed time. Concentric rings represent actor depth: human interaction at the center, then primary agent, subagents, and tool activity toward the outside. A radial sweep advances through the event history.

VISUAL ENCODING

Ring = actor depth • angle = time • arc length = duration • color = actor class • intensity = activity volume

KEY INTERACTIONS

Change time window and playback speed; isolate a ring; inspect a segment; toggle duration, event count, or token overlays; compare sessions in aligned layers.

DESIGN GUARDRAILS

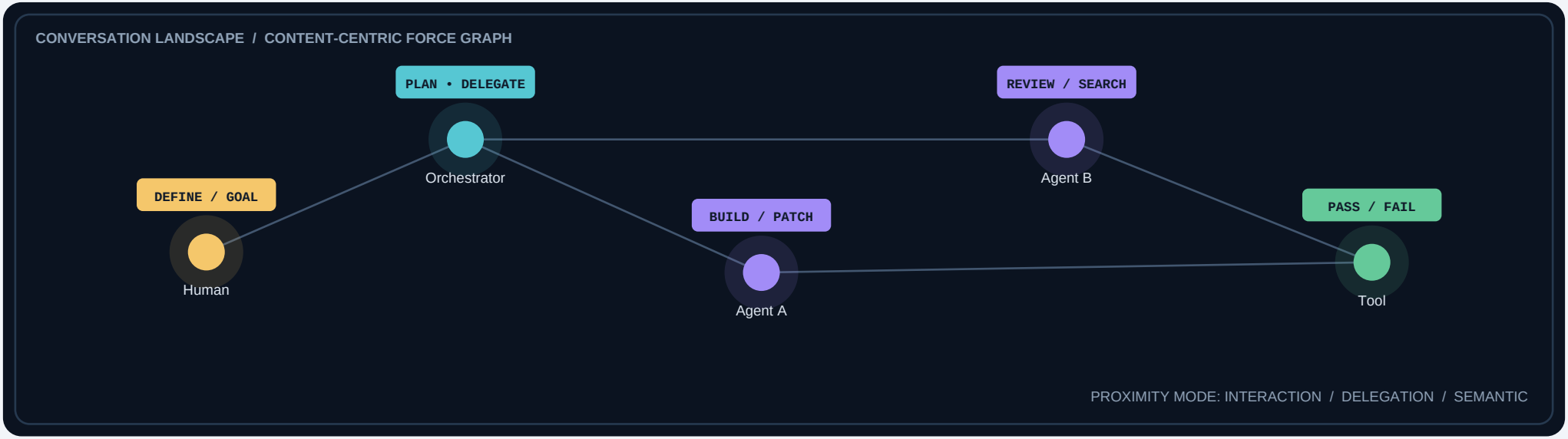
Keep actor depth and time visually distinct. Do not equate duration with computational work: waiting and active execution need separate encodings. Aggregate when actor counts grow.

PRIMARY QUESTION

When did activity happen, and how was it distributed?

Conversation Landscape

A content-centric force-directed map where words appear around the actors that emit them.



WHY THIS VIEW

Show not just that actors communicated, but what their communication was about. Turn conversation into a visible field of topics, delegation, and response.

VISUAL MODEL

Agents and tools are nodes in a force-directed graph. Short phrases or keyword clouds appear around the sending node. The recipient is drawn toward the delegating node; responses reinforce or reshape the relationship.

VISUAL ENCODING

Node = actor • nearby words = emitted content • attraction = interaction strength • word prominence = topic or message importance

KEY INTERACTIONS

Select a word to find source messages; expand a cloud to read the full message; switch interaction, delegation, and semantic layouts; replay evolution; filter by actor or topic.

DESIGN GUARDRAILS

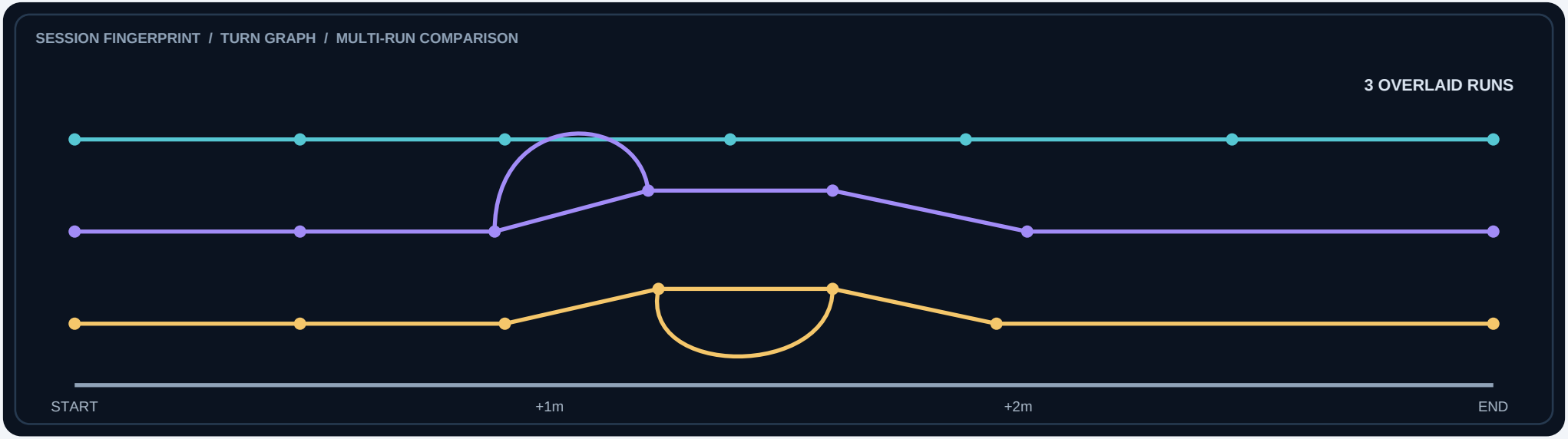
Make the force model explicit. Proximity means actual interaction only in interaction mode; semantic similarity belongs in a separate mode. Dampen motion and preserve original messages.

PRIMARY QUESTION

Who is communicating, and what are they communicating about?

Session Fingerprint

A compressed turn graph that makes the shape, timing, and resource profile of a run recognizable.



WHY THIS VIEW

Give each run a compact visual signature, then compare runs to discover differences in branching, retries, waiting, tool use, duration, and token expenditure.

VISUAL MODEL

The underlying structure is a turn/event graph. Nodes represent messages, tool calls, delegations, results, and terminal events; edges represent execution relationships. Branches reveal parallel work, loops reveal retries, and gaps or spans reveal waiting.

VISUAL ENCODING

Horizontal position = time • branches = delegated or parallel work • loops = retries • node size = tokens • color/shape = event type

KEY INTERACTIONS

Overlay sessions; switch absolute-time and start-aligned views; expand a branch; highlight common paths; filter by model, task, tool, or outcome; expand to the full turn graph.

DESIGN GUARDRAILS

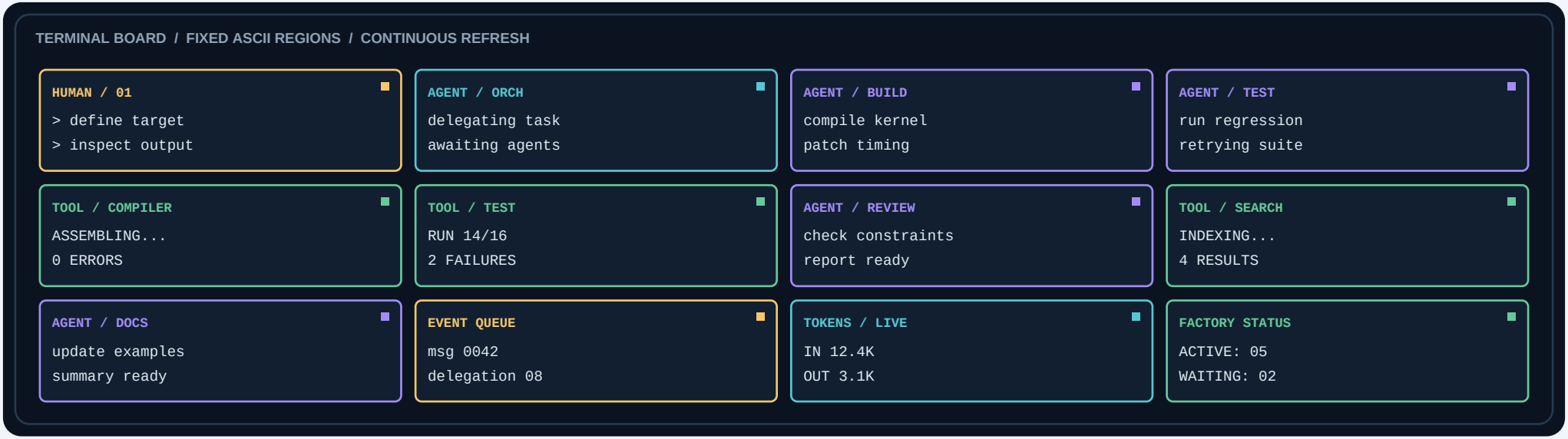
Absolute time reveals starts, ends, and overlap; normalized time reveals structural differences. Preserve links to original events. Collapse repeated motifs without hiding deviations.

PRIMARY QUESTION

What did a run look like, and how does it compare with others?

Terminal Board

A fixed ASCII grid of actor regions, constantly refreshed with words and short status fragments.



WHY THIS VIEW

Put communication itself on display. Like an airport or railway split-flap board, each actor has a known place and new information visibly replaces old information.

VISUAL MODEL

Every agent and tool is assigned a fixed rectangular region on a character grid. Words, phrases, status lines, and compact summaries emitted by that actor appear only in its region. New content flips, scrolls, or overwrites existing characters; the board geometry remains stable.

VISUAL ENCODING

Fixed region = actor identity • text = message or status • color = actor/event class • replacement animation = new event • optional indicators = queue, tokens, health

KEY INTERACTIONS

Configure grid and actor placement; choose raw fragments, keywords, or summaries; adjust refresh speed; pause/replay; inspect a region's history; switch themes.

DESIGN GUARDRAILS

Define deterministic behavior for simultaneous messages: queue, prioritize, or aggregate. Keep important content visible for a minimum duration. Treat the board as compressed view, not canonical transcript. Use true fixed-width character cells.

PRIMARY QUESTION

What are the actors saying or doing right now?